

- Simple Moving Average (SMA)
- Exponential Moving Average (EMA)
- Relative Strength Index (RSI)
- Bollinger Bands
- MACD (Moving Average Convergence Divergence)
- One indicator of your choice (Stochastic Oscillator, ATR, etc.)

2. Implement signal generation functions that use these indicators to generate trading signals:

- `check_ma_crossover()`: Detect when a short-term MA crosses a long-term MA
- `check_rsi_conditions()`: Identify overbought/oversold conditions
- `check_bollinger_band_breakout()`: Detect price breaking above/below bands
- `check_macd_signal()`: Identify when MACD crosses the signal line

3. Create portfolio and risk management functions:

- `calculate_position_size()`: Determine position size based on account size and risk percentage
- `calculate_stop_loss()`: Set stop-loss levels based on volatility or fixed percentage
- `track_portfolio_metrics()`: Calculate metrics like profit/loss and drawdown

4. Build a simple backtesting function that combines these components:

- Take historical price data as input
- Apply your technical indicators and signal generators
- Simulate trades based on the signals
- Track performance metrics